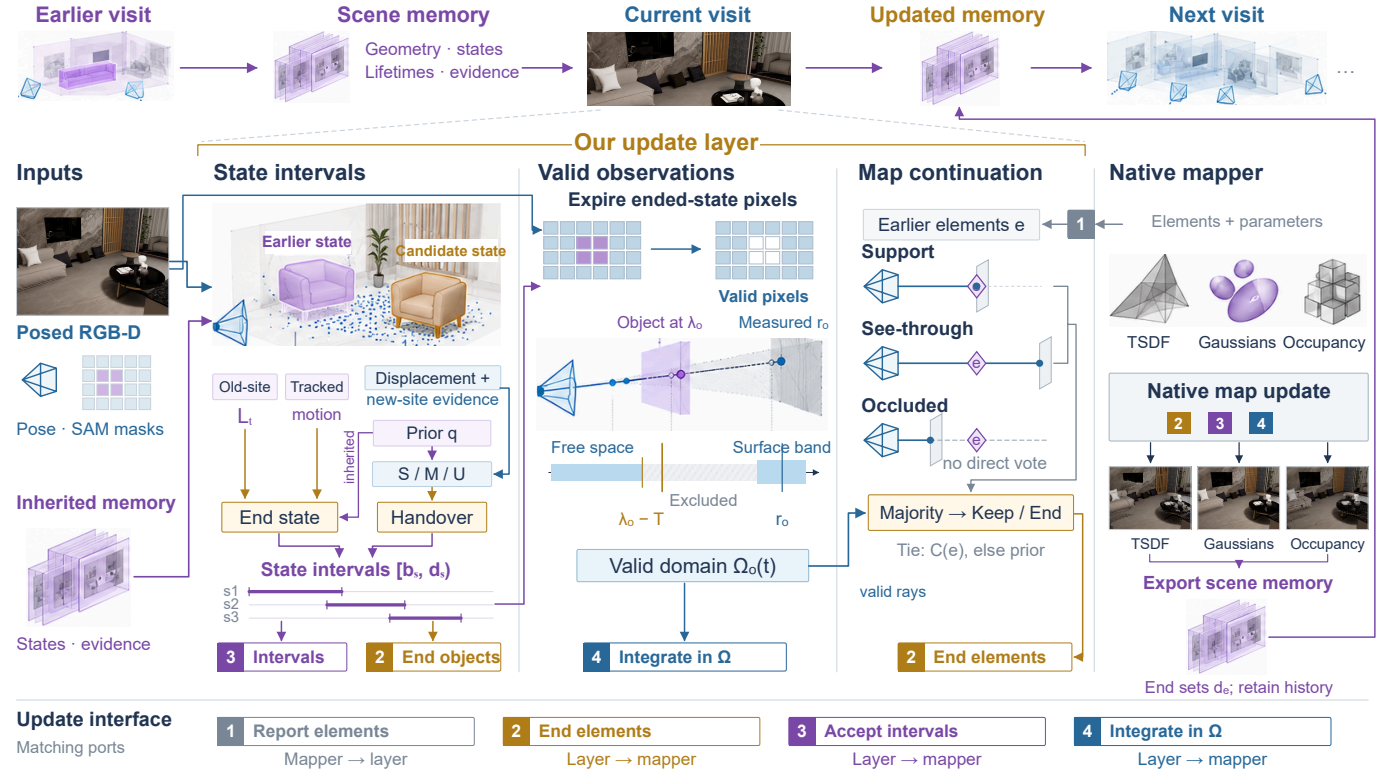

Earlier visit
Scene memory
Geometry · states
Lifetimes · evidence
Current visit
Updated memory
Next visit
Our update layer
Inputs
Posed RGB-D
Pose · SAM masks
Inherited memory
States · evidence
State intervals
Earlier state
Candidate state
Old-site
Tracked
motion
Displacement + new-site evidence
inherited
Prior q
S / M / U
End state
Handover
State intervals [bs, ds)
s1
s2
s3
3 Intervals
2 End objects
Valid observations
Expire ended-state pixels
Valid pixels
Object at λo
Measured ro
Free space
Surface band
Excluded
λo − T
ro
Valid domain Ωo(t)
4 Integrate in Ω
Map continuation
Earlier elements e
1
Support
See-through
Occluded
no direct vote
Majority → Keep / End
Tie: C(e), else prior
valid rays
2 End elements
Native mapper
Elements + parameters
TSDF
Gaussians
Occupancy
Native map update
2
3
4
TSDF
Gaussians
Occupancy
Export scene memory
End sets de; retain history
Update interface
Matching ports
1 Report elements
Mapper → layer
2 End elements
Layer → mapper
3 Accept intervals
Layer → mapper
4 Integrate in Ω
Layer → mapper